

MINEWATCH

End-to-End Deployment & Operations Guide

AI-Enabled Real-Time Mine Subsidence Monitoring,
Prediction and Early Warning System

SMART INDIA HACKATHON 2026 | PROBLEM STATEMENT SIH26025

Team	MineWatch
PS Category	Hardware
Deployment Model	Surface sensor nodes over/around underground mine panels
System Objective	Continuous sensing → communication → AI analysis → prediction → warning

Deployment guide for mine officers, field deployment personnel, control-room operators and system administrators.

1. Solution Overview

MineWatch is a distributed surface-monitoring system intended to continuously observe ground conditions above underground coal mine panels. The proposed solution combines multi-parameter sensor nodes, 915 MHz LoRa communication, gateway-side data handling, AI/ML analysis and a GIS-enabled dashboard for live monitoring and early warning.

The deployment model used in this guide assumes that the sensor nodes are installed on the surface at defined monitoring locations. Each node is associated with a mine, panel and physical surface location so that every measurement can be mapped back to its source.

1.1 End-to-End System

1. SENSE	2. IDENTIFY	3. TRANSMIT	4. PROCESS	5. VISUALIZE	6. WARN
Tilt Vibration Displacement Crack/strain	Node ID Location Panel	915 MHz LoRa → Gateway	Storage AI/ML Sensor fusion	GIS map Live values Trends	Risk status Alerts SOS

The submitted solution specifies BNO085 for tilt/inclination, ADXL355 for vibration, and two FBG sensors for displacement/strain and crack-related monitoring. Nodes communicate with a central gateway using 915 MHz LoRa. Results are intended to be presented through a GIS-enabled web/mobile dashboard with live readings, deformation trends, predicted risk zones and alerts.

1.2 Deployment Principle

- Every physical node receives a unique Node ID.
- Every Node ID is associated with a defined surface monitoring point.
- The node is electrically and functionally checked before being marked Active.
- LoRa connectivity is verified through the gateway before commissioning.
- The gateway forwards/stores data for server-side processing and dashboard visualization.
- The officer verifies the node on the GIS dashboard before completing deployment.

2. Roles and Responsibilities

Role	Primary Responsibility	When Used
Deployment Officer	Install, identify, power, test and commission surface nodes.	Field deployment
System Administrator	Register nodes, locations, gateways and authorized users.	Before/after deployment
Control-Room Officer	Monitor live conditions, node health, trends and alerts.	Routine operation
Maintenance Technician	Resolve sensor, power, node or communication faults.	Fault condition
Mine Management	Review risk information and operational reports.	Decision/support

3. Pre-Deployment Preparation

Before entering the field, the deployment team prepares the hardware, identities and deployment records.

3.1 Hardware Checklist

- STM32-based sensor node
- BNO085 tilt/inclination sensor
- ADXL355 accelerometer
- FBG sensors and optical components where required
- 915 MHz LoRa module

- Primary power supply and battery backup
- 3.3 V / 5 V regulation
- SOS/emergency button
- Protective enclosure and mounting hardware
- LoRa gateway and gateway power/network connection

3.2 Deployment Record

Prepare one record for each node containing:

- Node ID
- Mine
- Panel
- Surface monitoring location
- GIS coordinates
- Sensor configuration
- Assigned gateway
- Deployment date/time
- Deployment officer

3.3 Node Placement Layout and Inter-Node Spacing

Node positions are not selected arbitrarily. Each node location is fixed by a planned survey grid over the panel boundary, and the grid spacing is set so that every point on the panel surface remains within monitoring resolution and every node remains within reliable LoRa range of its gateway. This section gives the placement rule used to generate the grid, a worked reference layout with inter-node distances, and the field procedure for verifying that installed positions match the plan.

3.3.1 Placement Principle

- Node positions are laid out on a rectangular grid over the surface projection of the panel, aligned to the panel's long axis (strike) and short axis (dip), using the panel boundary coordinates from the mine survey plan as the reference frame.
- The grid origin (0 m, 0 m) is fixed at a known survey monument or a corner of the panel boundary so that every node position can be reproduced from the deployment record without re-surveying the whole panel.
- Row spacing and column spacing are equal by default, producing a square grid; unequal spacing is used only where the panel's aspect ratio or access constraints require it, and any deviation is recorded against the affected node IDs.
- Spacing is tightened near a goaf edge, fault, old workings boundary or any zone flagged high-risk in the geotechnical assessment, and relaxed only in zones confirmed low-risk by the same assessment.
- The gateway is placed as close to the geometric centre of the node grid as terrain and access allow, so that the longest node-to-gateway path is minimised.

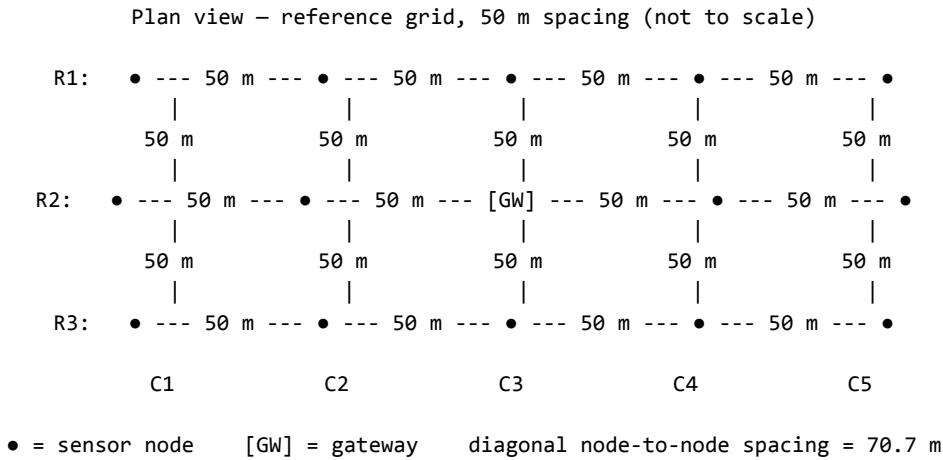
3.3.2 Standard Spacing Rule

Unless the site-specific geotechnical assessment specifies otherwise, apply the following default spacing:

- Normal-risk zone: 50 m spacing between adjacent nodes in both grid directions.
- Reinforced zone (within 50 m of a goaf edge, fault trace or known weak-roof boundary): 25 m spacing, i.e. one additional node inserted at the midpoint of each affected grid edge.
- Minimum of one node within 25 m of any active SOS-relevant structure (haul road, surface facility, access point) located above or adjacent to the panel.
- Maximum planned node-to-gateway straight-line distance: 500 m for open surface terrain with clear line of sight; reduce this figure where vegetation, stockpiles, buildings or terrain undulation will obstruct the LoRa path, and confirm the working figure with a link test before finalising node count.

3.3.3 Reference Grid Layout — Worked Example

The following layout illustrates the method for a reference panel measuring 200 m (strike) by 100 m (dip), using 50 m spacing. Apply the same method to the actual panel dimensions taken from the survey plan; node count and coordinates will differ accordingly.



Node IDs follow the pattern MW-R⟨row⟩C⟨column⟩. Rows are numbered from the panel boundary edge nearest the return airway (R1) toward the intake side (R3); columns are numbered from the survey origin along the strike direction (C1 to C5). The gateway mast is co-located with grid position R2-C3, at the centre of the layout.

Node ID	Grid Position	X-offset (m)	Y-offset (m)	Dist. to next node, same row (m)	Dist. to next node, same column (m)	Dist. to gateway (m)
MW-R1C1	R1-C1	0	0	50	50	111.8
MW-R1C2	R1-C2	50	0	50	50	70.7
MW-R1C3	R1-C3	100	0	50	50	50.0
MW-R1C4	R1-C4	150	0	50	50	70.7
MW-R1C5	R1-C5	200	0	—	50	111.8
MW-R2C1	R2-C1	0	50	50	50	100.0
MW-R2C2	R2-C2	50	50	50	50	50.0
MW-R2C3 (Gateway mast)	R2-C3	100	50	50	50	0.0
MW-R2C4	R2-C4	150	50	50	50	50.0
MW-R2C5	R2-C5	200	50	—	50	100.0
MW-R3C1	R3-C1	0	100	50	—	111.8
MW-R3C2	R3-C2	50	100	50	—	70.7
MW-R3C3	R3-C3	100	100	50	—	50.0
MW-R3C4	R3-C4	150	100	50	—	70.7
MW-R3C5	R3-C5	200	100	—	—	111.8

Distances in the table are straight-line (Euclidean) distances calculated from the planned grid coordinates.

Diagonal neighbour distance (e.g. R1-C1 to R2-C2) is 70.7 m throughout this layout and is not repeated in every row.

3.3.4 Gateway Placement and Link-Distance Check

- Place the gateway mast at, or as close as practicable to, the centre grid position so the longest node-to-gateway distance is minimised rather than left to the far edge of the panel.
- In the reference layout, the maximum planned node-to-gateway distance is 111.8 m (corner nodes R1-C1, R1-C5, R3-C1, R3-C5), well inside the 500 m planning limit given in 3.3.2.
- Planned distance is a starting point, not a substitute for a field link check. At each node, confirm RSSI and SNR at the gateway meet the system's minimum acceptable margin before commissioning; a short planned distance with an obstructed path can perform worse than a longer, clear one.
- Where a node fails the link check from its planned grid position, relocate it along the panel boundary to the nearest point with a clear path to the gateway, record the as-built offset from the planned position, and re-run the commissioning checklist in Section 5.

3.3.5 Field Verification of Node Positions

1. **Mark planned positions** — Before mobilising sensors, mark each planned grid position (survey stake or paint mark) using the offsets in the deployment record, working outward from the fixed grid origin.
2. **Survey the as-built position** — At each marked position, record the actual coordinates using handheld GNSS (± 3 m accuracy) or, where available, RTK GNSS or a total station tied to the mine survey control network (± 0.3 m accuracy).
3. **Check deviation against tolerance** — Compare the as-built coordinates to the planned grid coordinates. Accept the position if the deviation is within ± 3 m (handheld GNSS) or ± 0.3 m (RTK/total station); otherwise adjust the node position and re-survey.
4. **Record as-built values** — Enter the as-built coordinates, not the planned grid coordinates, into the deployment record and the GIS registration step (Field Deployment Procedure, step 9). The GIS map must reflect where the node actually is.
5. **Re-check inter-node spacing** — Confirm that no gap between adjacent as-built node positions exceeds the applicable spacing limit (50 m normal-risk, 25 m reinforced zone) given in 3.3.2; if it does, insert an additional node at the midpoint of the oversized gap before completing deployment.

4. Field Deployment Procedure

The following sequence is the standard field procedure from arrival at the monitoring point to dashboard verification.

1. **Receive the deployment assignment** — Confirm the mine, panel, monitoring area, number of nodes and assigned gateway.
2. **Locate the monitoring point** — Reach the surface location assigned to the node and verify that it matches the deployment record.
3. **Verify the node identity** — Check the physical Node ID label against the deployment record before installation.
4. **Install the node** — Securely mount the enclosure and install/connect the required sensors.
5. **Connect FBG sensing where applicable** — Install the optical/FBG sensing arrangement required for displacement, strain or crack-related monitoring.
6. **Power the node** — Connect the primary supply and verify that the battery-backup system is available.
7. **Run node diagnostics** — Check controller, BNO085, ADXL355, FBG subsystem where present, power and communication hardware.
8. **Establish LoRa communication** — Confirm that the node successfully transmits to the assigned 915 MHz LoRa gateway.
9. **Register the location** — Associate the Node ID with its mine, panel and GIS surface coordinates.
10. **Verify gateway reception** — Confirm that valid sensor packets are being received by the gateway.
11. **Verify server/data path** — Confirm that gateway data reaches the monitoring/data system and is available for processing.
12. **Commission the node** — Only after all checks pass, change the node status to Active.
13. **Verify on the dashboard** — Confirm the node appears at the correct map location and live data is visible.

5. Node Commissioning

A node must pass the commissioning checklist before it becomes part of the active monitoring network.

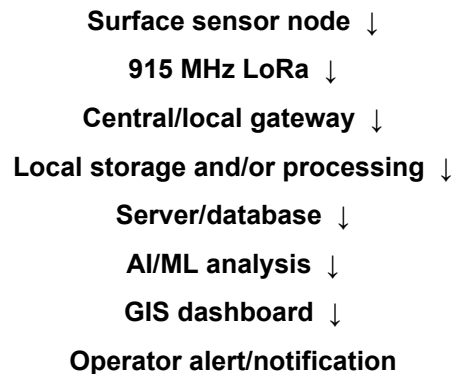
Check	Expected Result	Status
Node identity	Correct Node ID	PASS / FAIL
Power	Primary supply and backup available	PASS / FAIL
BNO085	Sensor responding	PASS / FAIL
ADXL355	Sensor responding	PASS / FAIL
FBG subsystem	Valid response where installed	PASS / FAIL / N/A
LoRa	Gateway receiving packets	PASS / FAIL

SOS	Button/event path available	PASS / FAIL
Location	Mine/panel/coordinates correct	PASS / FAIL
Server path	Data visible to processing system	PASS / FAIL
Dashboard	Node visible and mapped correctly	PASS / FAIL

Commissioning rule: A node is marked ACTIVE only after the physical, sensor, communication and dashboard checks have passed.

6. Gateway and Data Flow Verification

The gateway is the communication bridge between distributed surface nodes and the monitoring platform.



The proposed architecture also includes offline capability at the gateway: local storage/processing can maintain monitoring during internet loss and synchronize when connectivity returns.

7. Officer Dashboard Login and Verification

After field commissioning, the authorized officer moves to the MineWatch monitoring interface.

14. Open the MineWatch web/mobile dashboard from an authorized device.
15. Enter the assigned officer credentials.
16. Complete authentication and open the monitoring dashboard.
17. Confirm that the assigned mine and panel are visible.
18. Open the GIS map and locate the newly commissioned node.
19. Open the node detail view and verify live readings.
20. Check node health, battery and LoRa communication status.
21. Confirm that the node status is Active/Online.

7.1 Dashboard Sections

Section	Purpose
Overview	Overall node health, gateway status, current warnings and system condition.
GIS Map	Spatial location of nodes, deformation information and risk zones.
Nodes	Individual sensor readings, battery, communication and health status.
Alerts	Abnormal-condition warnings, SOS events and alert history.
Reports	Historical readings, deformation trends and system records.

8. Normal Monitoring Operation

Once commissioned, nodes operate continuously and the officer's role shifts from installation to monitoring.

- Check gateway and network health at the beginning of the shift.

- Check the number of Active/Online nodes.
- Review battery and sensor-health warnings.
- Review the GIS map for abnormal or changing areas.
- Review deformation and sensor trends rather than relying only on a single reading.
- Review generated warnings and notifications.
- Record and escalate events according to the mine's safety procedure.

9. Abnormal Condition and Alert Workflow

MineWatch is designed to combine tilt, vibration, displacement/strain and crack-related information with temporal and spatial analysis. An abnormal condition should therefore be treated as an event for verification and response rather than as an isolated sensor value.

22. Dashboard generates an abnormal-condition warning.
23. Officer opens the alert and identifies the affected node(s) and surface location.
24. Officer reviews current readings and historical trend.
25. Officer checks neighbouring nodes for spatial correlation.
26. Officer acknowledges and records the event according to site procedure.
27. Mine personnel perform the required field verification or safety action.
28. Event is retained in the monitoring history for subsequent analysis.

9.1 SOS Event

When the SOS button is activated, the node sends an emergency event containing the node identity/status and location through the monitoring system. The control-room officer uses the dashboard event to identify the affected surface location and follows the mine's emergency response procedure.

10. Communication or Internet Failure

A communication failure must be distinguished between a node-to-gateway failure and an internet/server connectivity failure.

Condition	Expected System Behaviour	Officer Action
Node ↔ Gateway lost	Node/gateway communication fault is indicated.	Check node power, LoRa link and physical installation.
Gateway ↔ Internet lost	Gateway continues local storage/processing where configured.	Monitor local gateway status; synchronize after connectivity returns.
Sensor fault	Node reports sensor/self-diagnostic fault.	Dispatch maintenance technician.
Low battery	Node health reports battery warning.	Schedule power/battery maintenance.

11. Deployment Completion Record

At the end of each deployment, the officer completes the commissioning record.

Mine: _____

Panel: _____

Deployment Area: _____

Node ID(s): _____

Gateway ID: _____

Deployment Date/Time: _____

Officer: _____

Number of Nodes Installed: _____

Number of Nodes Commissioned: _____

Number of Nodes Requiring Maintenance: _____

Dashboard Verification: PASS / FAIL

Deployment Notes: _____

12. End-to-End Operational Flow

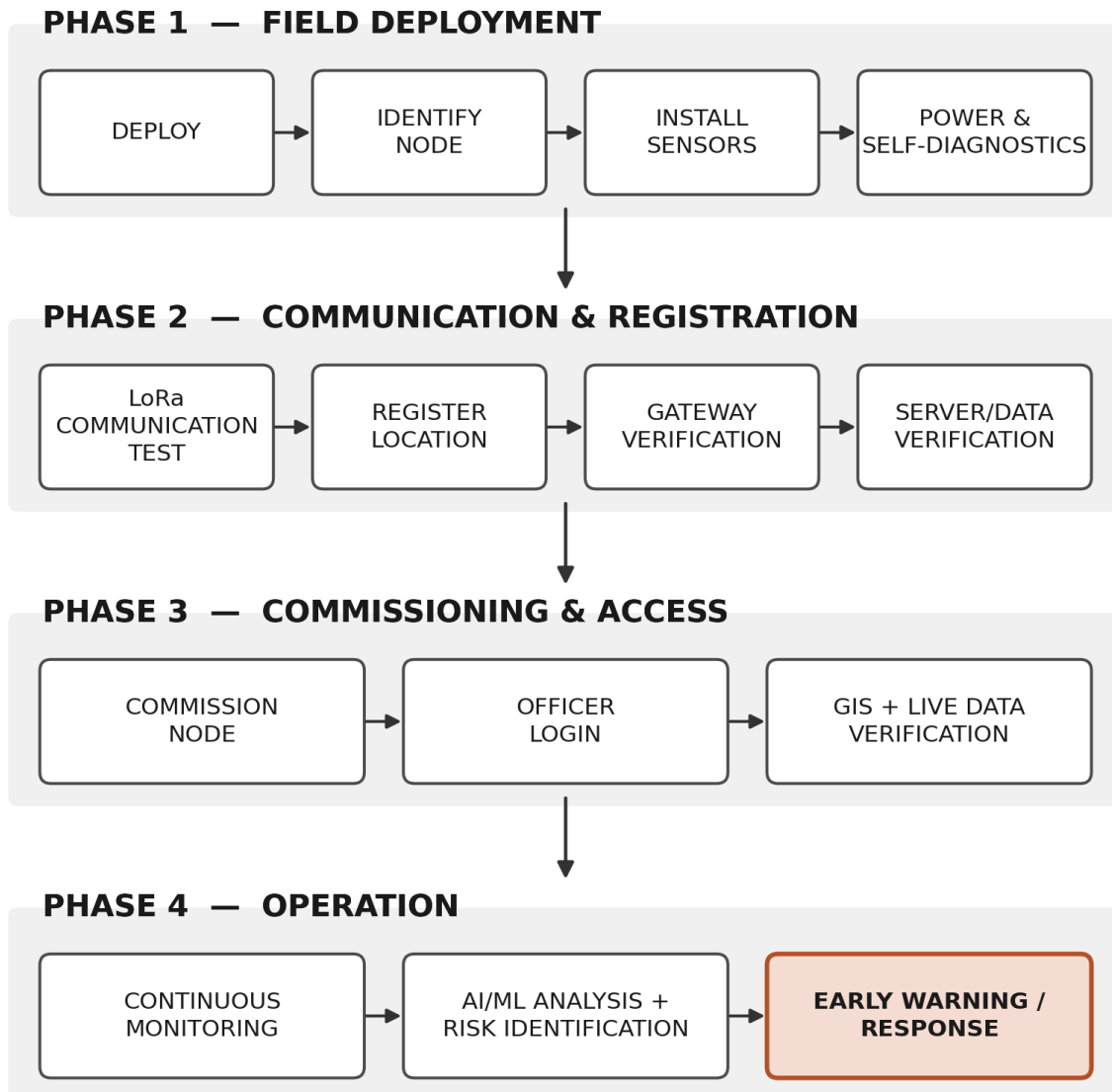


Figure 1. End-to-end operational flow, from field deployment to early warning and response.

13. Solution Coverage for SIH26025

This deployment workflow connects the physical field deployment directly to the intended MineWatch problem solution: distributed surface sensing of ground conditions, long-range wireless communication, data storage and analysis, AI/ML-based identification of abnormal deformation patterns, GIS-based visualization and early warning.

PS Requirement / Need	MineWatch Deployment Response
Continuous ground monitoring	Surface sensor nodes continuously collect tilt, vibration, displacement/strain and crack-related data.
Large-area deployment	Distributed nodes communicate over 915 MHz LoRa to a gateway.
Early detection	AI/ML analysis uses real-time and historical sensor information to identify abnormal patterns.
Spatial understanding	GIS associates each measurement with its surface monitoring location and displays risk zones.
Operational reliability	Battery backup, self-diagnostics and gateway-side offline capability support continued operation.
Emergency support	SOS events can be transmitted with node status and location.
Scalability	The architecture is intended to scale from sensor → panel → mine → coalfield.

SOURCE AND SCOPE

Prepared from the MineWatch SIH 2026 submission for Problem Statement SIH26025. The submitted material defines the sensing architecture, LoRa communication, gateway/data flow, AI/ML analysis, GIS dashboard, self-diagnostics, battery backup and SOS functions. Specific authentication screens, database schemas, exact dashboard controls and deployment-site safety procedures are implementation details and should be finalized by the project/system team and mine operator.